

Optimization-Based Design of Organ Allocation Policy: Mitigating Biological Access Disparities in Lung Transplantation

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Abbreviations:

HLA, human leukocyte antibodies; HRSA, Health Resources and Services Administration; OPTN, Organ Procurement and Transplantation Network; STAR, Standard Transplant Analysis and Research; WLAUC, Waitlist area under the curve

Abstract

Patient attributes such as blood type and chest cavity size impact access to lung transplantation, increasing waitlist mortality. The transplant community seeks to mitigate the adverse impacts of biologic characteristics on a patient's access to transplantation. However, organ allocation is shaped by complex interactions, and policymakers often must rely on expert intuition to translate their objectives into policy. In this work, we apply a new computational technique to design a modification to current lung allocation policy optimized to mitigate adverse effects of patient biologic characteristics. This method leverages high-speed simulation to iteratively refine scoring rules while capturing the complexity of the transplant system. Under simulation, the proposed adjustments reduce range in median waiting time by 17% (95% CI: 10–24%) and in transplant rate by 28% (95% CI: 18–38%) across blood type and chest cavity size groups, with minimal effects on other metrics. This new computational technique which we call rating scale optimization can help policymakers close the critical gap between the principles that guide organ allocation policy and the reality experienced by patients.

1. Introduction

Current lung allocation policy in the United States reflects a preference among members of the transplant community that two primary factors should most strongly influence a patient's access to donated organs: medical urgency and post-transplant survival.¹ Additionally, the community advocated to provide expanded access for pediatric patients, as well as those who were previous organs donors themselves. The continuous allocation score, implemented in March, 2023, reflects these preferences.² However, in practice, other biologic factors such as a patient's ABO blood type, chest cavity size, and sensitization to human leukocyte antibodies (HLA) antigens exert undesired influence on access to lung transplantation^{3,4}. Lyden et al⁵ report that in recent years patients with blood type O had transplant rates 37% lower than those with

blood type A or B, that patients below the 25th percentile in height had transplant rates 32% lower than those above the 75th percentile, and that patients with a cPRA of at least 50% had transplant rates 27% lower than those with a cPRA of 0%. While 14% of patients that listed during 2024 had blood type O and an effective height (defined in the next section) less than 158 cm, these patients made up 27% of those that died or became too sick to receive a transplant that year. Together, these differences underscore a critical gap between the principles guiding lung allocation policy and the reality experienced by patients.

The challenge is that there exists substantial variation among patients in the size of their potential donor pool and the amount of competition they face for their donor pool. While patients with blood type O are only compatible with blood type O donors and compete with patients of all other blood types, patients with blood type AB are compatible with donors of any blood type and compete with fewer patients for donors with blood type A, B and AB. A similar situation arises with chest cavity size. In lung transplantation, it is preferable for the chest cavities of the recipient and donor to be matched in size.⁶ However, compared to donors, lung transplant patients tend to have smaller chest cavities. They have a shorter average height and often have restrictive lung diseases that further reduce their chest cavity size. This means that for patients with a smaller chest cavity, there are fewer appropriate donors compared to patients with a larger chest cavity.

Though variation in patients' biologic characteristics drives variation in experienced supply and demand, it is possible to compensate for this by granting patients with more restrictive requirements greater priority for those organs that they are compatible with. Lung allocation policy has long included these sorts of mechanisms. However, the level of priority granted may be too much or too little and policymakers have had limited analytical tools to determine the

appropriate amount. The historical differences in access mentioned above attest to the difficulty in successfully fine-tuning these mechanisms using earlier methods for policy development.

There have been several approaches taken to account for biological factors in lung allocation. Under the initial continuous distribution lung allocation policy, patients were assigned values p_{abo} , p_{cPRA} , and p_{chest} that estimated the fraction of donors they would be incompatible with, given their blood type, cPRA, or height and diagnosis group.^{2,7} The function $(100^p - 1)/99$ was used to convert these values into three separate subscores. This approach does not capture the compounding effects that patients with multiple biological disadvantages face. To address this, Rose et al⁸ developed a single value to capture a patient's compatible donor supply based on their blood type, height, and diagnosis group. This was converted to a subscore with the reciprocal function $1/x$. While these approaches represent important progress, their scoring functions were defined through intuition, and therefore may not optimally address differences in access. Also, these approaches only capture variation in donor supply. They did not consider the competition that patients face for their donor supply.

In collaboration with the Organ Procurement and Transplantation Network (OPTN) Lung Transplantation Committee, we sought to modify current policy to better compensate for a patient's blood type and chest cavity size. In this work, we present a method we call *rating scale optimization* which employs new high-speed simulation software and iterative optimization algorithms to support the policy design process. Rather than relying on pre-specified functional forms, flexible, non-parametric scores are learned from repeated simulation to optimally adjust for different biologic characteristics. This method was applied with the aim of designing an allocation policy that reduced the difference in transplant access across patient groups.

2. Materials and Methods

2.1 Historical Data and Analysis of Access Trends

This study used data from the OPTN covering all donors, wait-listed candidates, and transplant recipients in the US, submitted by the members of the OPTN. The Health Resources and Services Administration (HRSA), U.S. Department of Health and Human Services provides oversight to the activities of the OPTN contractor.

Information on historical patients, organ donors, and lung transplants were from the OPTN Standard Transplant Analysis and Research (STAR) dataset. Custom analytic files were used to supplement STAR data and included information about patient listing centers, donor hospitals, waitlist history, program status data and uncut match run history data for implementation and verification of historical allocation policy.

Effective height was created as a proxy for chest cavity size, since direct measurements of patient chest cavity size were unavailable and the underlying lung disease diagnosis can impact chest cavity size. The effective height h_e of a patient of height HGT in diagnosis group DGN was conceived to be the height of a healthy person with a similarly sized chest cavity. In this way, patients with similar chest cavity sizes but different heights and diagnosis groups could be grouped by their effective height. Effective height was defined as

$$h_e(\text{HGT}, \text{DGN}) = \begin{cases} \text{HGT} + 5.08 \text{ cm}, & \text{if DGN is Group A,} \\ \text{HGT} - 2.54 \text{ cm}, & \text{if DGN is Group B,} \\ \text{HGT} + 2.54 \text{ cm}, & \text{if DGN is Group C,} \\ \text{HGT} - 7.62 \text{ cm}, & \text{if DGN is Group D,} \end{cases}$$

Hence, the chest size of a 165 cm patient with diagnosis in Group D was considered equal to the chest size of a 157.38 cm healthy person or a 152.3 cm patient with a Group A diagnosis. These

adjustments were selected by the OPTN Lung Transplantation Committee and based on Rose et al⁸.

The metrics used for measuring transplant access, defined in Table 1, were transplant percentage, WLAUC at transplant, waiting time at transplant, and transplant rate. For most analyses, patients were grouped by 4 blood types (O, A, B, and AB) and 5 intervals of effective height (≤ 158 cm, 158–165 cm, 165–170.1 cm, 170.1–177.7 cm, and >177.7 cm), which led to a total of 20 patient groups. Variation in transplant access was quantified using range and group-size-weighted standard deviation across groups. Weighted standard deviation considers the number of patients in each group and is larger if more populous groups are on the extremes. Range is agnostic to group size.

Table 1: Definitions of transplant access metrics.

Metric	Definition
Transplant Percentage	An estimate of the percentage of patients, potentially of a particular type, that will receive a transplant. This is computed as the percentage of patients removed from the waiting list during a particular period of time that were removed due to receipt of a transplant.
WLAUC at Transplant	“Waitlist Area under the [survival] curve,” or, patient’s estimated days of survival within 365 days without a transplant (according to the model used for the Medical Urgency attribute listed in Table 2) at the time they received a transplant. For a group of patients, a distribution of this metric can be computed from transplant recipients in the group.
Waiting Time at Transplant	The number of days since a patient registered on the waiting list at the time they received a transplant. For a group of patients, a distribution of this metric can be computed from transplant recipients in the group.
Transplant Rate	The number of transplants received by a group of patients during a particular period of time divided by the aggregate time the patients were listed on the waiting list.

2.2 Historical Lung Allocation Policies

Since March 9, 2023 OPTN lung allocation policies have employed a policy framework called *continuous distribution*. Under this framework eligible transplant patients are ranked by a

score known as the *composite allocation score*, which is a weighted average of several subscores. Each subscore, or *attribute*, uses a formula called a *rating scale* to capture a reason for prioritizing a patient. Rating scales assign a score between 0 and 1 based on certain patient and donor characteristics. The fraction of the overall composite score that is determined by any one attribute comes from the attribute's weight in the weighted average of all attributes.

At the time of writing, three continuous distribution lung allocation policies have been implemented by the OPTN. Table 2 summarizes the attributes, weights, and rating scales across these policies. The initial continuous distribution policy, implemented on March 9, 2023, featured nine attributes, including Medical Urgency, Post-Transplant Outcomes, and Blood Type. The September 27, 2023 update modified the rating scale of the Blood Type attribute, keeping the attributes and weights unchanged. A third policy, implemented on May 7, 2026, merged the Proximity Efficiency and Travel Efficiency attributes into a single Placement Efficiency attribute, increasing the importance of patient-donor proximity. We refer to the May 7, 2026 policy as “current” policy and the September 27, 2023 policy as “historical” policy since simulations were performed over 2024.

Table 2: Definitions of recent and proposed lung allocation policies.

WLAUC is the patient's expected days of survival within 365 days without a lung transplant and PTAUC is their expected days of survival within 1,826 days (5 years) of receiving a lung transplant. DIST is the great-circle distance in nautical miles between the patient's transplant center and the donor's hospital. The value p_{CPRA} is the estimated fraction of donors with at least one HLA antigen for which the patient has pre-formed antibodies, the value p_{chest} is the estimated fraction of donors whose lungs would be mis-sized for the patient based on the patient's height and diagnosis group. The values ABO, HGT, DGN are the patient's blood type, height, and diagnosis group. The function h_e computes a proxy for chest cavity size we call effective height based on height and diagnosis group. The function f compensates for a patient's blood type and chest cavity size. Both h_e and f are defined in the main text. Full models for WLAUC and PTAUC and the look-up table for p_{chest} is provided in the OPTN policy book effective March 9th, 2023. Formulas for the remaining functions f_{TE}^1 , f_{PE}^1 , f_{PE}^2 are provided in the supplemental materials.

Attribute	Weight	Rating Scale Formula
March 9 th , 2023 Lung Allocation Policy		
Medical Urgency	0.250	$= (25^{1-\text{WLAUC}/365} - 1)/24$
Transplant Outcome	0.250	$= \text{PTAUC}/1826$

Pediatric Access	0.200	= 1 if patient age < 18 at listing, 0 otherwise
Prior Living Donor Access	0.050	= 1 if prior living donor, 0 otherwise
Travel Efficiency	0.050	= $f_{TE}^1(\text{DIST})$
Proximity Efficiency	0.050	= $f_{PE}^1(\text{DIST})$
cPRA	0.050	= $(100^{p_{cPRA}} - 1)/99$
Chest Cavity Size	0.050	= $(100^{p_{chest}} - 1)/99$
Blood Type	0.050	= 0.091 if O, 0.009 if A, 0.049 if B, 0 if AB
September 27 th , 2023 Lung Allocation Policy		
Medical Urgency	0.250	= $(25^{1-WLAUC/365} - 1)/24$
Transplant Outcome	0.250	= PTAUC/1826
Pediatric Access	0.200	= 1 if patient age < 18 at listing, 0 otherwise
Prior Living Donor Access	0.050	= 1 if prior living donor, 0 otherwise
Travel Efficiency	0.050	= $f_{TE}^1(\text{DIST})$
Proximity Efficiency	0.050	= $f_{PE}^1(\text{DIST})$
cPRA	0.050	= $(100^{p_{cPRA}} - 1)/99$
Chest Cavity Size	0.050	= $(100^{p_{chest}} - 1)/99$
Blood Type	0.050	= 1 if O, 0.061 if A, 0.448 if B, 0 if AB.
May 7 th , 2026 Lung Allocation Policy		
Medical Urgency	0.236	= $(25^{1-WLAUC/365} - 1)/24$
Transplant Outcome	0.236	= PTAUC/1826,
Pediatric Access	0.189	= 1 if patient age < 18 at listing, 0 otherwise
Prior Living Donor Access	0.047	= 1 if prior living donor, 0 otherwise
Proximity Efficiency	0.150	= $f_{PE}^2(\text{DIST})$
cPRA	0.047	= $(100^{p_{cPRA}} - 1)/99$
Chest Cavity Size	0.047	= $(100^{p_{chest}} - 1)/99$
Blood Type	0.047	= 1 if O, 0.061 if A, 0.448 if B, 0 if AB.
Proposed Lung Allocation Policy		
Medical Urgency	0.236	= $(25^{1-WLAUC/365} - 1)/24$
Transplant Outcome	0.236	= PTAUC/1826,
Pediatric Access	0.189	= 1 if patient age < 18 at listing, 0 otherwise
Prior Living Donor Access	0.047	= 1 if prior living donor, 0 otherwise
Proximity Efficiency	0.150	= $f_{PE}^2(\text{DIST})$
Biological Disadvantages	0.142	= $\min(1, \frac{1}{3}(100^{p_{cpra}} - 1)/99 + f(\text{ABO}, h_e(\text{HGT}, \text{DGN}))$

2.3 Rating Scale Optimization

Rating scale optimization is a computational procedure for the design of rating scales.

Policymakers select optimization metrics and objectives and decide what information the rating scale will use to assign prioritization scores. Then an initial policy is simulated and the results of the simulation are analyzed. An automated algorithm makes modifications to the rating scale

being optimized to improve its performance. In this work, this process was repeated for 100 iterations, and the best-performing iteration was selected. Then, the selected rating scale was simplified by approximating it with piecewise-linear functions.

The policies simulated during the optimization procedure were based on the May 7, 2026 policy. The policies included the same attributes and rating scales, except the separate Blood Type and Height attributes were replaced with one combined attribute with a weight of 0.15 and the weights of the other attributes (Medical Urgency, Transplant Outcome, Pediatric Access, Prior Living Donor Access, Proximity Efficiency, and cPRA) were scaled from their values under the approved policy so that they would add to the remaining weight of 0.85.

The rating scale of the combined attribute was the portion of the policy being optimized. During optimization iteration $k = 1, \dots, K$, the rating scale of the combined Chest Cavity Size-Blood Type attribute was defined non-parametrically, as a look-up table based on a patient's blood type and effective height. Similar to current OPTN policy, effective height was rounded to the nearest centimeter (between 63 and 223) when performing a look up. We denote this look-up table function f_k .

The initial rating scale function f_1 assigned scores by adding the Blood Type and Chest Cavity Size rating scales. At each iteration k , the policy using f_k was simulated 10 times. These simulation results were then used to compute the next function f_{k+1} . Specifically, for each blood type and effective height, the mean waiting time at transplant was computed among the 15 recipients of nearest effective height, plus any additional recipients with an effective height within 2.54 cm. The mean across all blood types and effective heights was computed to be a reference point for equalization. The score assigned for a particular blood type and effective height was increased (decreased) if mean waiting time at transplant was less (greater) than the

overall average by an amount proportional to the difference. Between every iteration, the rating scale was smoothed with a standard Gaussian, then clamped between 0 and 1. The cPRA rating scale was not included in the optimization process due to historical inconsistent reporting of lung transplant patient’s unacceptable HLA antigens.

Upon termination of the optimization procedure at iteration K , the non-parametric function f_K was approximated using the piecewise-linear function f presented in the results. To achieve greater access equalization without altering attribute weights, the proposed policy also merged the cPRA, Blood Type, and Chest Cavity Size attributes into one Biological Disadvantages attribute. This allows a patient’s blood type and chest cavity size to determine a larger fraction of their composite allocation score. See Table 2 for the attributes and rating scales of this proposed policy.

2.4 Simulation Model

In this study, policy evaluation relied on computer simulation of the lung transplant system. The model used in this work followed similar principles to those used by the OPTN for policymaking in the past. Simulations involved two phases: a *waitlist simulation* and an *allocation simulation*. They were performed using an algorithm recently developed by our group. As discussed in Pivo et al⁹, this algorithm was measured to reduce the time required to run a kidney and pancreas allocation simulation from nearly 7 hr down to about 15 s. While the algorithm provides a more efficient way to perform a simulation, the modeling principles remain similar to those used in earlier OPTN simulation studies. Further details of the model structure and training processes can be found in the supplement.

Simulation verification and validation was performed with a layered approach. Extensive unit tests were created to verify proper behavior of low-level functions as well as of integrated components. Additionally, simulator-generated lung matches were compared directly to those generated by the UNOS computer system during 2024 to ensure accuracy. Finally, simulations were performed under historical policy and simulation outcomes were compared to historical system data collected by the OPTN. Metrics that were compared include: transplant percentage, WLAUC at transplant, waiting time at transplant, transplant rate, donor-recipient distance, graft failures, transplant centers at top of match run, PTAUC at transplant, and donor age. These metrics are reported for lung transplant patients overall and stratified by age, cPRA, blood type and effective height, OPTN region, sex, and diagnosis group when applicable.

3. Results

3.1 Historical Trends in Lung Transplant Access

OPTN data was analyzed to assess historical levels of variation in lung transplant access due to blood type and chest cavity size. Figure 1 shows four measures of access across different blood type-effective height patient groups from 2015 through 2024. Gray shading shows the range in access across these groups. Over this ten-year period, the average range in transplant percentage was 23.3%, the average range in median WLAUC at transplant was 78.4 days, the average range in median waiting time at transplant was 58.4 days, and the average range in transplant rate was 651 transplants per hundred years. In 2024, the most recent year analyzed, the range in transplant Percentage was 15%, the range in median WLAUC at transplant was 91.4 days, the range in median waiting time at transplant was 42.8 days, and the range in transplant rate was 1,967 transplants per hundred years.

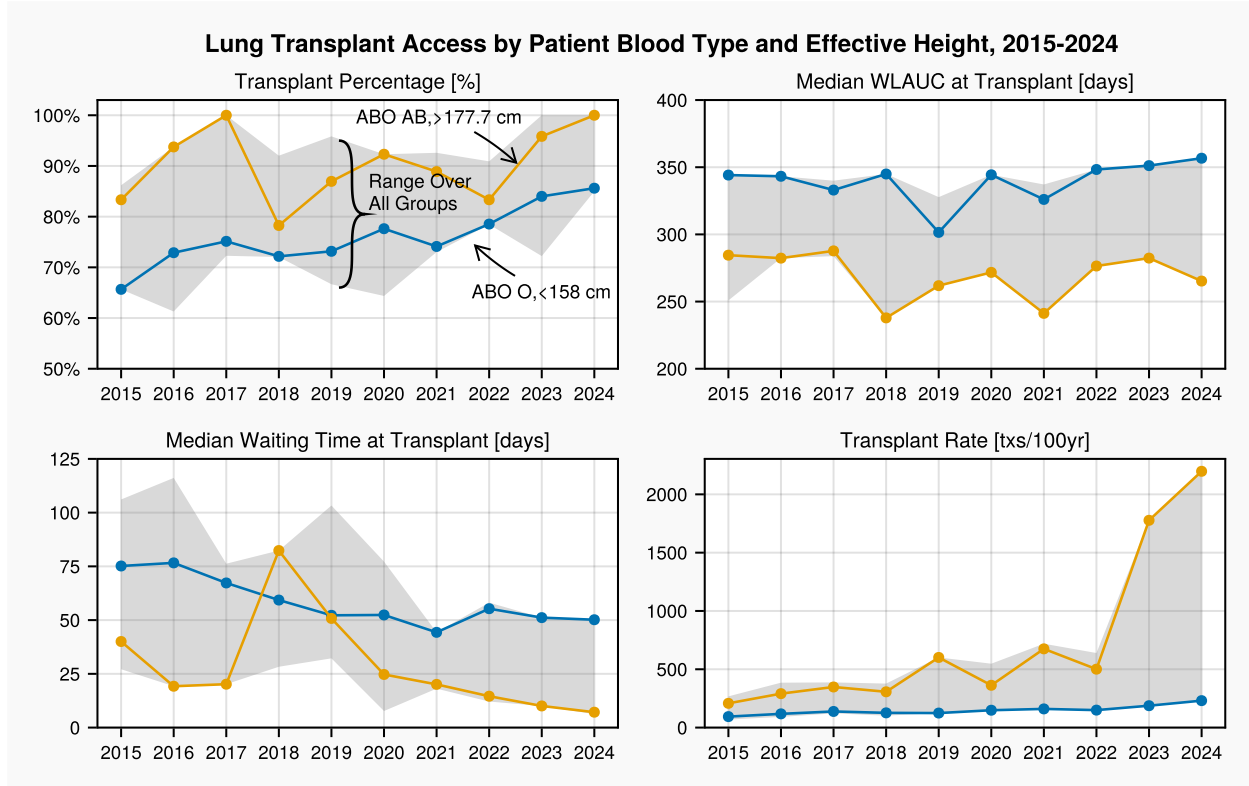


Figure 1: Historical variation in access to lung transplantation by blood type and chest cavity size from 2015 to 2024.

Patients were stratified by their blood type and chest cavity size, as measured by effective height. Four measures of transplant access are shown.

3.2 Simulation Validation

The simulation model used in this work was validated by performing simulations of lung allocation in 2024 and comparing these simulation results to historical data obtained from the OPTN. Included here, Figure 2 shows lung transplant counts and waitlist mortality for lung patients overall and for various patient stratifications. Validation results for other metrics referenced in this work are provided in the Supplemental Materials. Simulation results closely reflected historical data.



Figure 2: Select validation results for the simulation model.

(a) Comparison of number of lung transplants performed historically in 2024 (blue) and under 15 simulations of 2024 (yellow) over all and by select stratifications. (b) Comparison of waitlist deaths in 2024 (blue) and under 15 simulations of 2024 (yellow) overall and by select stratifications. The range bars run from the minimum to maximum simulated value over 15 runs.

3.3 Rating Scale Optimization

Results of the rating scale optimization procedure are depicted in Figure 3. Several iterates of the scoring function f_k are shown in Panel A. The first version, based on existing policy, assigned scores that were largely symmetric with respect to effective height, for instance with patients receiving the same score if their effective height was 3 cm above or 3 cm below 167.64 cm. After optimization, the symmetry of f_1 was lost. Instead, higher scores were granted to patients with smaller effective heights. In addition, scores assigned to blood type O and B patients were increased relative to scores assigned to blood type A and AB patients. The rating scale selected at this stage was from iteration 76.

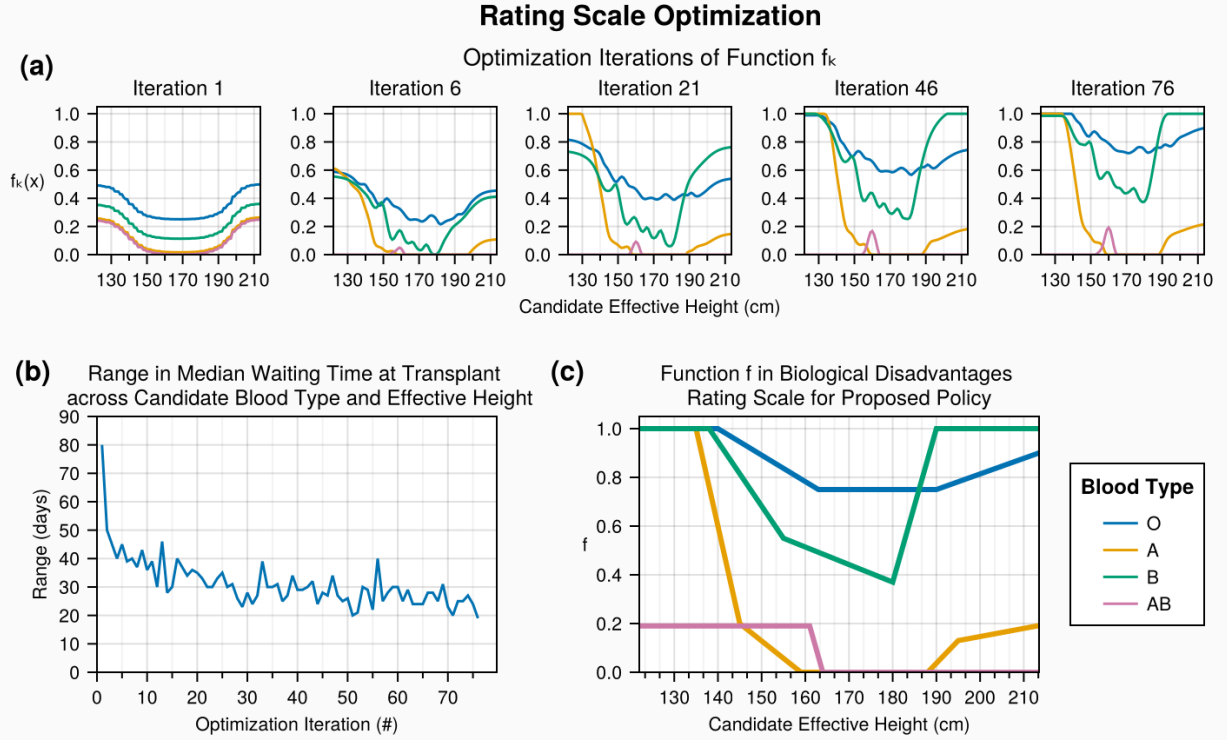


Figure 3: Optimization of a biological disadvantages rating scale for lung allocation policy.

An initial policy based on the one approved by the OPTN in November 2025 was simulated and then adjusted over many iterations to reduce variation in access to lung transplantation due to blood type and chest cavity size. (a) Five iterates of the blood type-chest cavity size scoring function f_k . (b) Reduction in variation in transplant access during the optimization process, as measured by the range in Median Waiting Time at Transplant across blood type and effective height groups. (c) The final function f used in the proposed policy, which is a piecewise-linear approximation of the final iterate f_{76} .

Panel B shows how the range in Median Waiting Time at Transplant across patient blood types and effective height groups changed over the course of the optimization process. There was a rapid decrease over the first few iterations, followed by a slower decline until termination of the process. The algorithm exhibited a tendency towards functions that oscillated with respect to effective height. This may arise from misalignment between how access was measured and how access was altered. These oscillations were dampened through gaussian filtering of the functions at each iteration.

Upon completing the optimization process, the raw function f_{76} was approximated with piecewise-linear functions to produce the function used in the proposed policy. This final function f is shown in Panel C. The largest modification was to the scoring function for blood type AB patients. In f_{76} , blood type O, A, and B patients were assigned maximal scores to patients with effective heights less than 137.16 cm. This characteristic was added for blood type AB patients in f .

3.4 Evaluation of Proposed Policy

In this section, we compare simulation results for the current and proposed policies. The proposed policy is based on the current policy, but adjusts the mechanism used to compensate for patient cPRA, blood type, height, and diagnosis group. Both policies are defined in Table 2. The cPRA, Chest Cavity Size, and Blood Type attributes are merged into one Biological Disadvantages attribute. This new attribute uses the optimized blood type and chest cavity size scoring function, f from the previous section. First, the policies are compared in terms of variation in lung transplant access across blood type and effective height groups. Then the underlying factors driving the improvements are analyzed. Finally, the effects on other metrics are considered.

In Figure 4, we compare the current and proposed policies with respect to variation in access to lung transplantation associated with patient blood type and chest cavity size. Each policy was simulated 50 times. The top panel shows the range and weighted standard deviation in four metrics across the 20 groups under the current and proposed policy. Weighted standard deviation of these four metrics decreased 22% (95% CI: 18–26%), 18% (95% CI: 15–20%), 38% (95% CI: 34–41%), and 36% (95% CI: 32–40%), indicating more equalized access. The ranges in these metrics all decreased, except for transplant percentage, which increased by 2% (95% CI: –7–

12%). The bottom panel, shows results for blood type O patients with an effective height less than 158 cm. This group has the most restrictive blood type and chest cavity size. Under the proposed policy, the group's transplant percentage increased by 4% (95% CI: 3–4%), median WLAUC at transplant increased by 4% (95% CI: 4–5%), their median waiting time at transplant decreased by 25% (95% CI: 22–29%), and their transplant rate increased by 46% (95% CI: 43–49%).

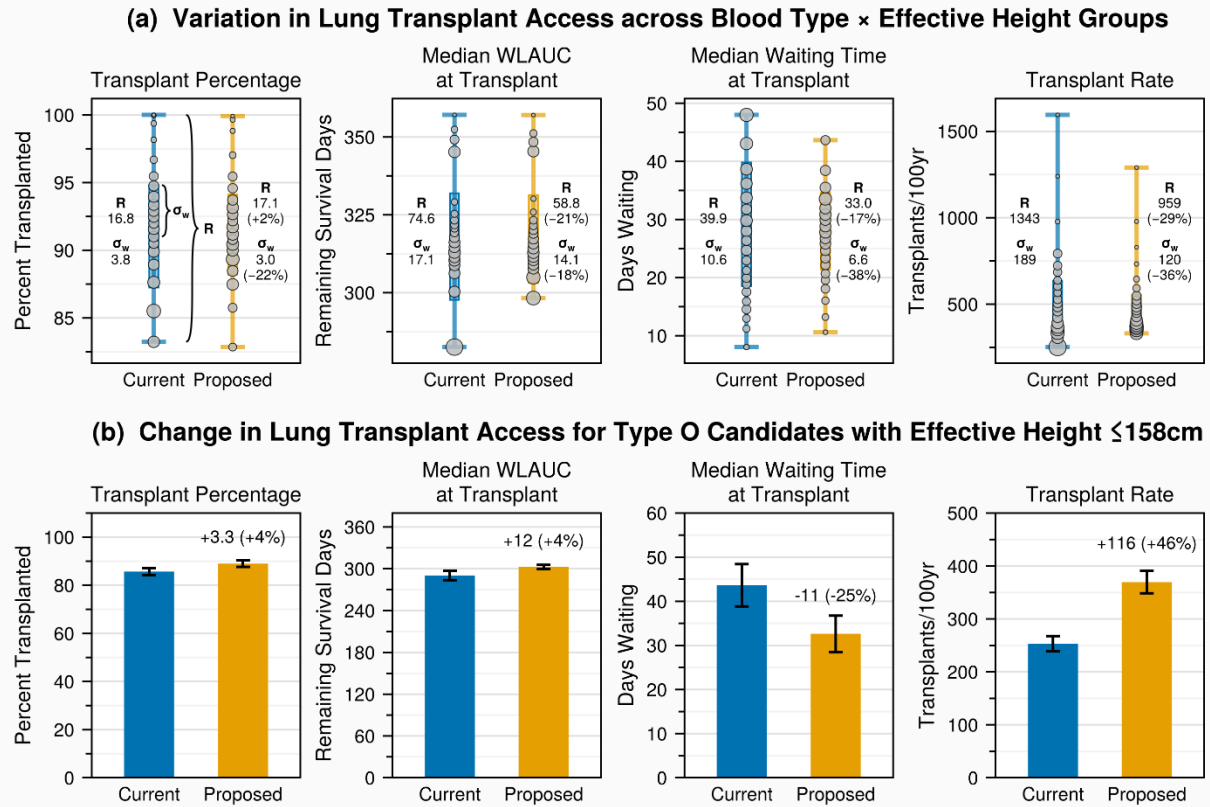


Figure 4: Variation in lung transplant access across patient groups under the approved and proposed allocation policy.

Lung allocation in 2024 was simulated under two policies: the policy approved in November 2025 and a proposed policy using the optimized biological disadvantages rating scale. Four access metrics are compared: transplant percentage, median urgency (WLAUC) at transplant, median waiting time at transplant, and transplant rate (per 100 patient listing years). **(a)** Spread of each metric across blood-type × effective-height groups, averaged over 50 simulations. In each simulation, the groups present under both policies were ranked from highest to lowest on the corresponding metric; points show the metric value at each rank, averaged across simulations, with point area proportional to the mean patient count occupying that rank (common scale across all four panels). The whisker (labeled R) is the across-group range (max – min), computed per simulation and averaged over simulations. The box (labeled σ_w) is the patient-count-weighted standard deviation across groups within a simulation, likewise averaged over simulations. Labeled percent changes compare the proposed policy to the approved policy. The proposed policy narrows both the range and the weighted spread for all four metrics. **(b)** Access gains under the proposed policy for the historically most disadvantaged group, ABO O patients with effective height ≤158 cm. Bars show the mean value across simulations for each policy; error bars show the across-simulation standard deviation. Labeled values are the absolute and percent change from the approved to the proposed policy.

Further, the proposed policy achieved similar performance to current policy in “off-target” metrics, such as overall number of lung transplants, and waitlist removals due to death or becoming too sick (Figure 5).

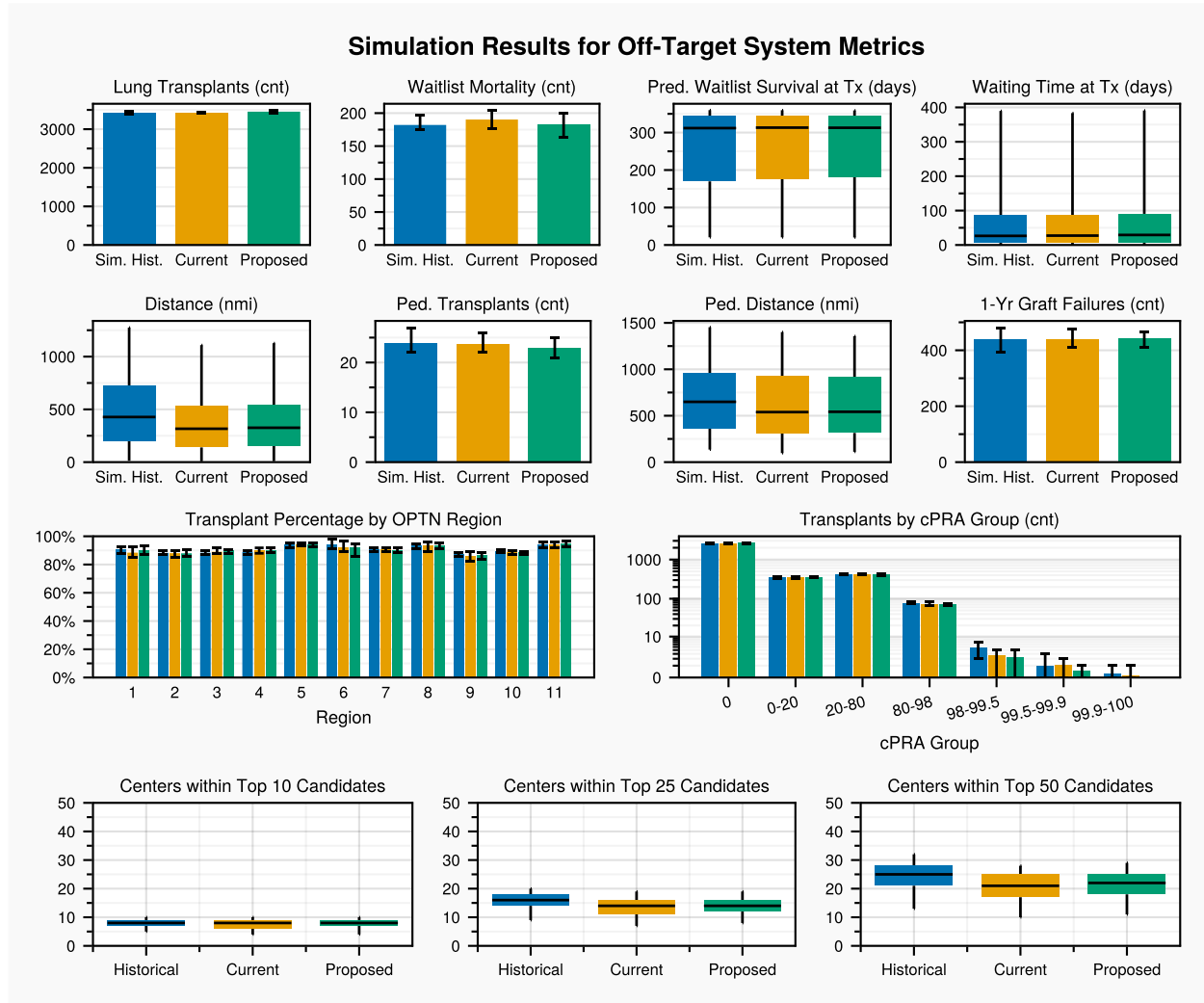


Figure 5: Simulation results for off-target system metrics.

A collection of important metrics that policymakers did not want to impact substantially through this policy adjustment. No significant differences are observed between the proposed and approved policies in terms of lung transplants performed, waitlist mortality, predicted waitlist survival at transplant, waiting time at transplant, distance, pediatric transplants performed, distance for pediatric transplants, 1-year graft failures, transplant percentage by OPTN region, transplants by cPRA, or the number of centers appearing at the top of match runs. The vertical axis of the cPRA plot uses a pseudolog scale to display values spread across several orders of magnitude.

3.5 Effect of Proposed Policy on Match Runs and Allocation Patterns

Changes in transplant access are achieved through changing the prioritization of patients on match runs, so the rate at which patients appeared in the top three sequence numbers of match runs was measured across patient groups. Figure 6 shows the ratio of these rates between the

current and proposed policies. For the ≤ 158 cm, blood type O group, this rate was 41% higher under the proposed policy, and for the 165–170.1 cm, blood type AB group, this rate was 54% lower.

**Ratio of Frequency of a Top 3 Ranking under
Proposed and Current Policy**

		Patient Blood Type			
		O	A	B	AB
Patient Effective Height	≤ 158 cm	1.94	1.01	0.88	1.76
	158-165cm	0.98	0.58	0.53	1.96
	165-170.1cm	1.02	0.37	1.75	0.58
	170.1-177.7cm	0.74	0.7	0.66	0.91
	>177.7 cm	1.6	0.87	1.15	1.05

Figure 6: Ratio of frequency of a top 3 ranking by patient blood type and effective height under proposed and current policy.

Frequency of top 3 rankings for a particular patient group was computed as the average number of times a patient in that group was ranked within the top 3 patients of a match run per active time on the waiting list. For example, blood type O patients with an effective height less than 158 cm were within the top 3 patients on a match run 1.94x more often under the proposed policy than under the current policy.

The proposed policy adjusts how lungs from donors of different blood types and chest sizes are allocated to patients of different blood types and chest sizes. Figure 7 compares these allocation patterns under the current and proposed policy. Under the proposed policy, more blood type O and B organs are allocated to patients with the same blood type, and fewer organs from smaller-sized donors are allocated to taller patients.

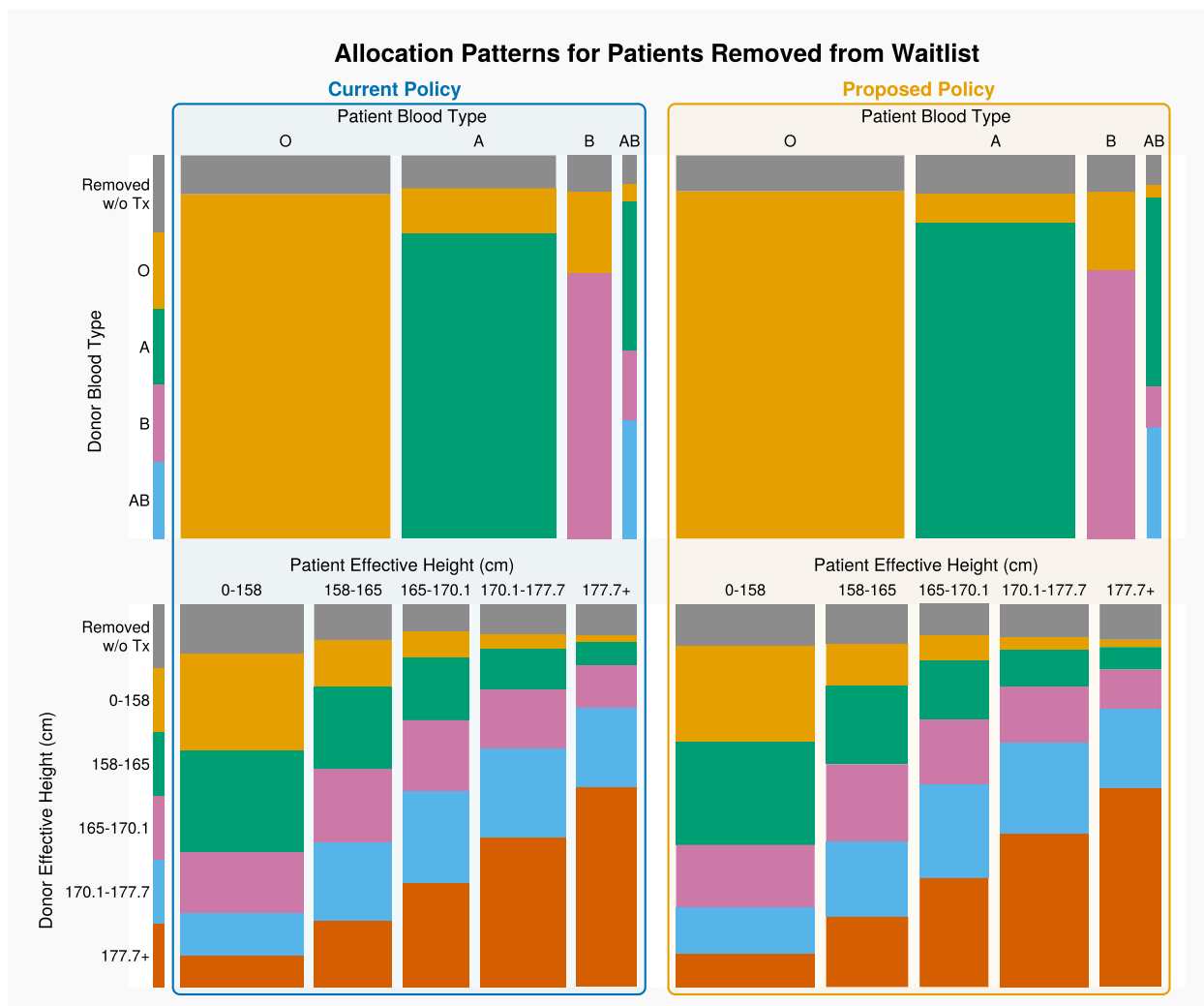


Figure 7: Allocation patterns for lung transplant patients removed from the waiting list.

The bar widths in each of these 4 mosaic plots are proportional to the number of patients of a particular type removed from the waiting list during simulation. The height of each tile represents the fraction of these patients that were removed either without a transplant or with a transplant from a donor of a particular blood type or effective height. Compared to the current policy, the proposed policy reserves more blood type O donors from blood type O patients and more donors with smaller effective heights for patients with smaller effective heights.

4. Discussion

In this work, we introduced a new computational method we call rating scale optimization to support the development of organ allocation policy. Rating scale optimization provides a method for determining the scores that should be assigned to patients to achieve a specified policy objective. The method iteratively improves flexible, non-parametric scoring functions over many rounds of simulation. By leveraging detailed simulation models, the method accounts for complex allocation dynamics in the organ transplant system. Recently developed high-speed simulation software has made this approach computationally tractable. Rating scale optimization provides a broadly applicable and principled approach for addressing complex allocation challenges in the U.S. organ transplant system. We demonstrated the method by designing a targeted modification of current lung allocation policy to reduce adverse impacts of a patient's blood type and chest cavity size on their access to lung transplantation.

Historically, access to lung transplantation has varied substantially based on a patient's blood type and chest cavity size (Figure 1). It is possible to compensate for these biologic factors by granting patients with more restrictive requirements greater priority for the organs that they are compatible with. However, the level of priority granted must be carefully tuned. Policymakers have sought to compensate for a patient's biologic characteristics in earlier allocation policies, but determining appropriate priority levels has been challenging, as evidenced by the historical variation in access.

Rating scale optimization was employed to design a new compensatory scoring function based on a patient's blood type, height, and diagnosis group (Figure 3). A proposed policy was defined by modifying current policy to use this new scoring function in place of the current ABO and Height rating scales. Transplant access metrics were selected to compare these policies in

collaboration with the OPTN Lung Transplantation Committee. These metrics captured a hierarchy of access concerns. The likelihood of receiving a transplant, medical severity at transplant, waiting time at transplant, and transplants received per time waiting were captured by transplant percentage, median WLAUC at transplant, median waiting time at transplant, and transplant rate, respectively.

Simulation results found that variation in lung transplant access due to blood type and chest cavity size, as measured by weighted standard deviation (σ_w), decreased in all four access metrics (Figure 4). Measured by range, which does not consider the size of the patient groups, variation decreased in all access metrics except for transplant percentage. High variance in transplant percentage of small groups (<20 patients), such as type AB patients with effective height 158 – 165 cm, limited potential improvement in transplant percentage range. These results were achieved without significant changes in other important metrics (Figure 5).

This work comes with several limitations. First, unreported unacceptable antigens mean that patient cPRA may not accurately reflect a patient’s immunosensitivity. For this reason, compensation for cPRA scores was not included in the optimization process. Second, rating scale optimization is performed with respect to a model of the transplant system. Inaccuracies in the model can affect the performance of the policy upon implementation. Offer acceptance models may fail to predict behavioral changes, out-of-sequence allocation can impact model accuracy, and post-transplant outcome and waitlist continuation models are impacted by selection bias in training data. Nevertheless, validation results provide evidence that the simulation model can faithfully capture much of the complexity of the system. Third, this work only considered policies that compensate for patient chest cavity size without taking donor chest cavity size into account. This means that smaller-sized patients will be prioritized even for larger donors. This

may result in additional declined offers during the allocation process. This can be mitigated by updating patients' donor criteria. Nonetheless, exploring policies that consider patient and donor size together is a potential direction of future research.

These results represent an important step towards alleviating a longstanding challenge in lung transplantation, one that has impacted thousands of patients seeking lifesaving treatment. The method demonstrated in this paper can also address a variety of other challenges facing the organ transplant system. These advances can aid policymakers and improve care for the many patients in need of an organ transplant in the United States.

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Disclosure

The authors of this manuscript have no conflicts of interest to disclose as described by the *American Journal of Transplantation*.

Data Availability Statement

The data that support the findings of this study were provided by the Organ Procurement and Transplantation Network (OPTN) under a Data Use Agreement. Restrictions apply to the availability of these data, which were used under license for the current study and are therefore not publicly available. Data are available from the OPTN (<https://optn.transplant.hrsa.gov>) upon submission of a data request and execution of a data use agreement, subject to OPTN's standard approval process. The simulation results supporting the findings of this study are available from the authors upon reasonable request.

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Supplemental Materials

1. Distance-Based Rating Scale Definitions

Table 2 of the main text references three functions, f_{TE}^1 , f_{PE}^1 , and f_{PE}^2 , used in OPTN lung allocation policy to assign priority scores based on donor-patient proximity. Definitions of these functions were omitted from the main text for brevity. In the following definitions, DIST denotes the great-circle distance in nautical miles between the patient's transplant center and the donor's hospital. First, the travel efficiency rating scale is defined as

$$f_{TE}^1(\text{DIST}) = \begin{cases} 1 - \frac{6.3\text{DIST}}{116989.1}, & \text{DIST} \leq 43.44, \\ 1 - \frac{253.93\text{DIST} - 10757.0472}{116989.1}, & 43.44 < \text{DIST} \leq 67.17, \\ 1 - \frac{149.49\text{DIST} - 3741.8124}{116989.1}, & 67.17 < \text{DIST} \leq 86.9, \\ 1 - \frac{21.15\text{DIST} + 7410.9336}{116989.1}, & 86.9 < \text{DIST} \leq 5181, \\ 0, & 5181 < \text{DIST}. \end{cases}$$

Next, the first proximity efficiency rating scale is defined as

$$f_{PE}^1(\text{DIST}) = \begin{cases} 1, & \text{DIST} \leq 45, \\ 1 - 0.15 \frac{\text{DIST} - 45}{45}, & 45 < \text{DIST} < 90, \\ \frac{0.875}{1 + \exp(0.0025\text{DIST} - 3.75)}, & \text{DIST} \geq 90. \end{cases}$$

Finally, the second proximity efficiency rating scale is defined as

$$f_{PE}^2(\text{DIST}) = \begin{cases} 1 - 0.0004\text{DIST}, & \text{DIST} \leq 50, \\ 1.08 - 0.002\text{DIST}, & 50 < \text{DIST} \leq 100, \\ 0.9 - 0.0002\text{DIST}, & 100 < \text{DIST} \leq 150, \\ 0.885 - 0.0001\text{DIST}, & 150 < \text{DIST} \leq 250, \\ \frac{263}{300} - \frac{1}{15000}\text{DIST}, & 250 < \text{DIST} \leq 400, \\ 1.25 - 0.001\text{DIST}, & 400 < \text{DIST} \leq 1000, \\ \frac{5181}{16736} - \frac{1}{16736}\text{DIST}, & 1000 < \text{DIST} \leq 5181, \\ 0, & 5181 < \text{DIST}. \end{cases}$$

These rating scales are shown in Figure S1.

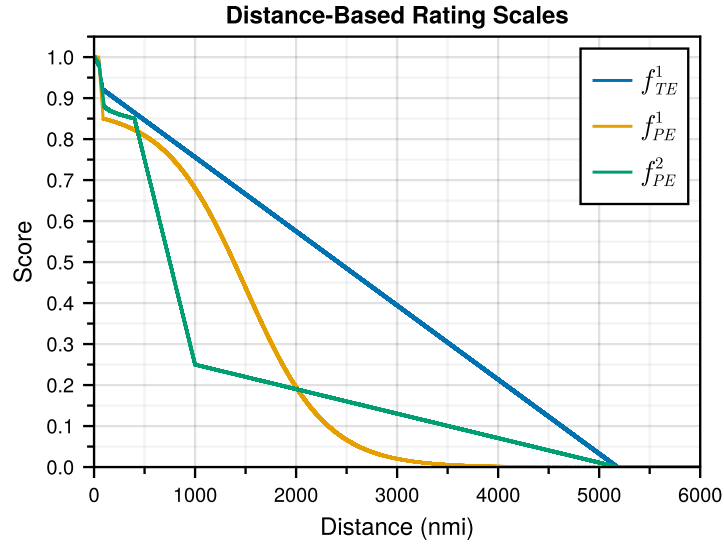


Figure S1. Distance-based rating scales used in OPTN lung allocation policy.

2. Simulation Model Details

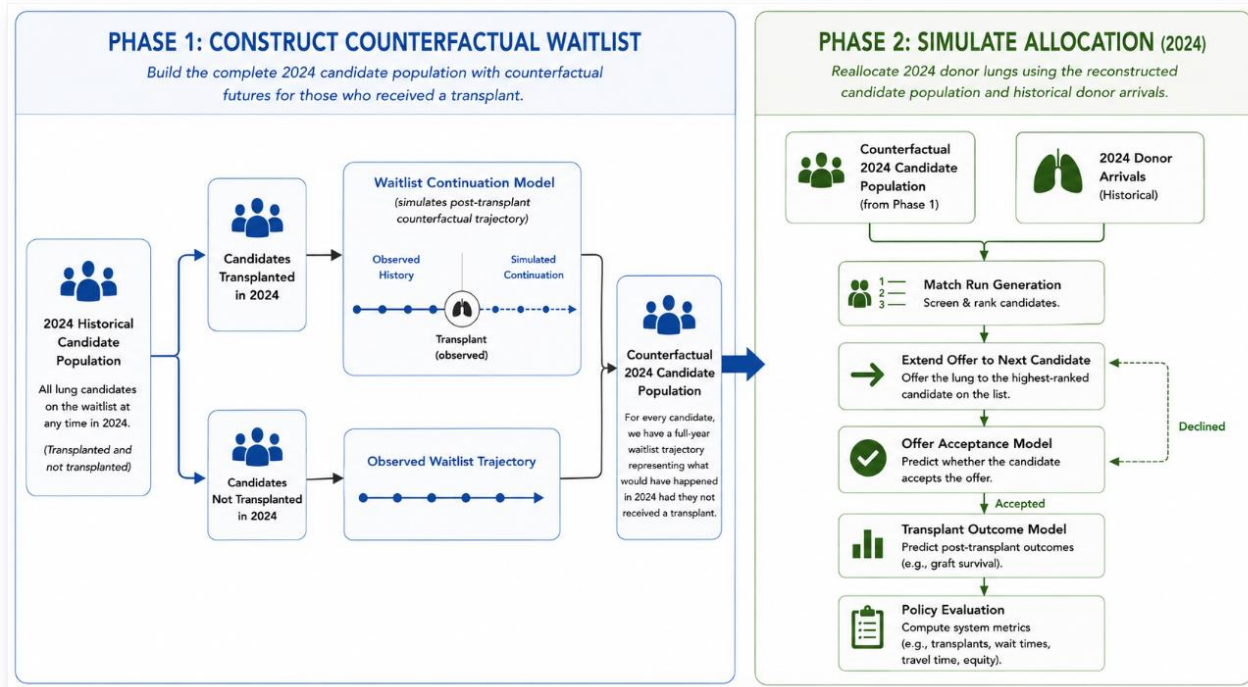


Figure S2: Structure of the simulation model.

The simulation proceeds in two phases. First, in the waitlist simulation phase, a waiting list is created from historical patient data. For historical lung transplant recipients, a clinical trajectory is simulated for the period of time after they received their transplant predicting what would have happened if they had not received their transplant and instead remained on the waiting list. Second, in the allocation simulation

phase, the lungs historically transplanted in 2024 are reallocated under the lung allocation policy under study. Simulated offers are made and evaluated with a probabilistic acceptance model trained on historical offer acceptance data. A transplant outcome model predicts years of survival for transplant recipients

Simulations involve predicting how patients and the waitlist would evolve without transplants being performed. For some patients, their complete clinical trajectory during 2024 was captured by historical waitlist data and was used without modification. These were the patients that either were removed from the waiting list without receiving a transplant during 2024 or those that had not yet received a transplant by the end of the year. However for patients that received a transplant during 2024, their clinical trajectory without transplantation during the simulation period is unobserved. A *waitlist continuation model* was used to simulate the clinical trajectory for historical lung transplant recipients under the counterfactual scenario in which they do not receive a transplant. This model used a Cox-Proportional Hazards model to predict time until waitlist removal, and a 3-class, multinomial logistic regression model to predict whether removal was due to waitlist death, becoming too sick for transplantation, or a different reason. The features used in the model were age, bilirubin, BMI, ventilation status, creatinine, presence of a restrictive lung disease, need of assistance with daily living, level of oxygen they are receiving, mean pulmonary arterial pressure, six minute walk test, weight, and number of days on the waitlist.

The model was trained on historical patients registered for a lung transplant during the period January 1, 2015 to December 31, 2024. Each patient was treated as a single observation, where a random time was sampled between their registration and removal which determined the time at which the features were measured, and served as the reference for time until removal. Patients that were removed due to transplant, or still on the waitlist by January 1, 2025 were considered censored.

A second clinical model was employed during allocation simulation, the *transplant outcome model*. This model determined how long a patient would survive if a patient received a particular transplant in simulation. This model was a Cox-Proportional Hazards model trained on patients who received a lung transplant during the period January 1, 2015 to December 31, 2024. The features used in this model included donor attributes (age, sex, blood type, height, weight, BMI, creatinine, diabetes, mechanism of death, DCD donor), patient attributes (diagnosis, sex, blood type, age, time on the waitlist), and the distance between the donor hospital and transplant center. Recipients that were still alive on January 1, 2025 were considered censored.

The final model was the *transplant offer acceptance model*. The simulator reallocated the lungs donated for transplant in 2024 by screening patients for each donor according to OPTN lung eligibility criteria and then ranking the patients according to the allocation policy under study. The acceptance model determines the probability that a patient accepts the left, right or both lungs when extended a transplant offer. The model first computed a probability of acceptance for an available left lung using a logistic regression model. Then, acceptance of the left lung was determined. The availability and acceptance of the left lung were passed as additional features to a second logistic regression model to compute the probability of accepting the right lung. The features used in this model included donor attributes (age, blood type, DCD/DBD, height, weight, BMI, BSA, peak creatinine, peak SGOT, peak sodium, and final creatinine), patient attributes (lung laterality preferences, age, diagnosis group, cPRA, sex, blood type, and

transplant center), as well as distance between the donor hospital and transplant center, offer number, and pTLC ratio.

The logistic regression parameters were fit using lung allocation data from February 1, 2024 to April 30, 2024. The lung transplant recipients contained in STAR file data were identified within the historical OPTN match run of their donor. Donors with multiple lung match runs were excluded to avoid ambiguity in which match run was used for allocation. Any patient prioritized in the portion of a match where lungs were available were considered to have declined any available lung they did not receive in a transplant and accepted a lung they did receive.

3. Additional Simulator Validation Plots

The main text includes validation results for simulated Transplant Counts and Waitlist Mortality for various patient stratifications and overall. Validation results for the other metrics referenced in the text are provided here. Validation results for the four transplant access metrics (transplant percentage, WLAUC at transplant, waiting time at transplant, and transplant rate) are included first. These are followed by plots for donor-recipient distance, graft failures, transplant centers at top of match run, PTAUC at transplant, and donor age.

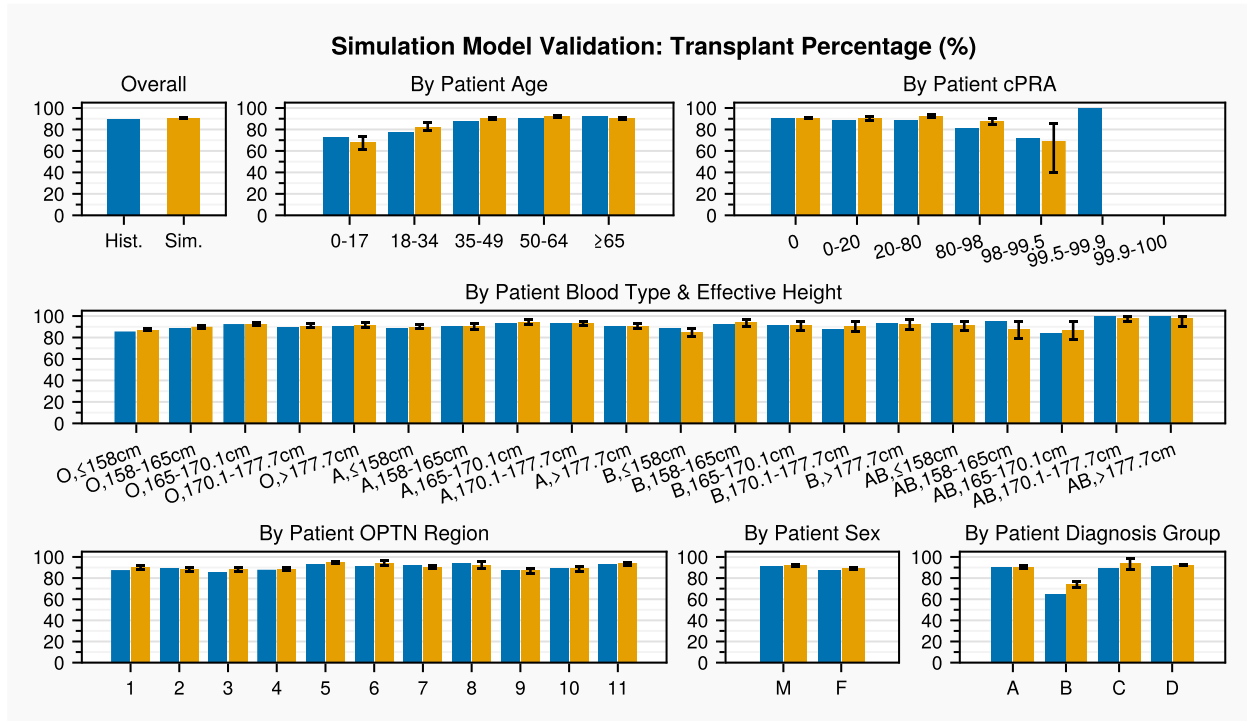


Figure S2. Comparison of transplant percentage computed from 2024 data (blue) and from 15 simulations of 2024 (yellow) over all patients and select stratifications. The range bars run from the minimum to maximum simulated value. No bar is displayed if no patients within the associated group were removed from the waiting list.

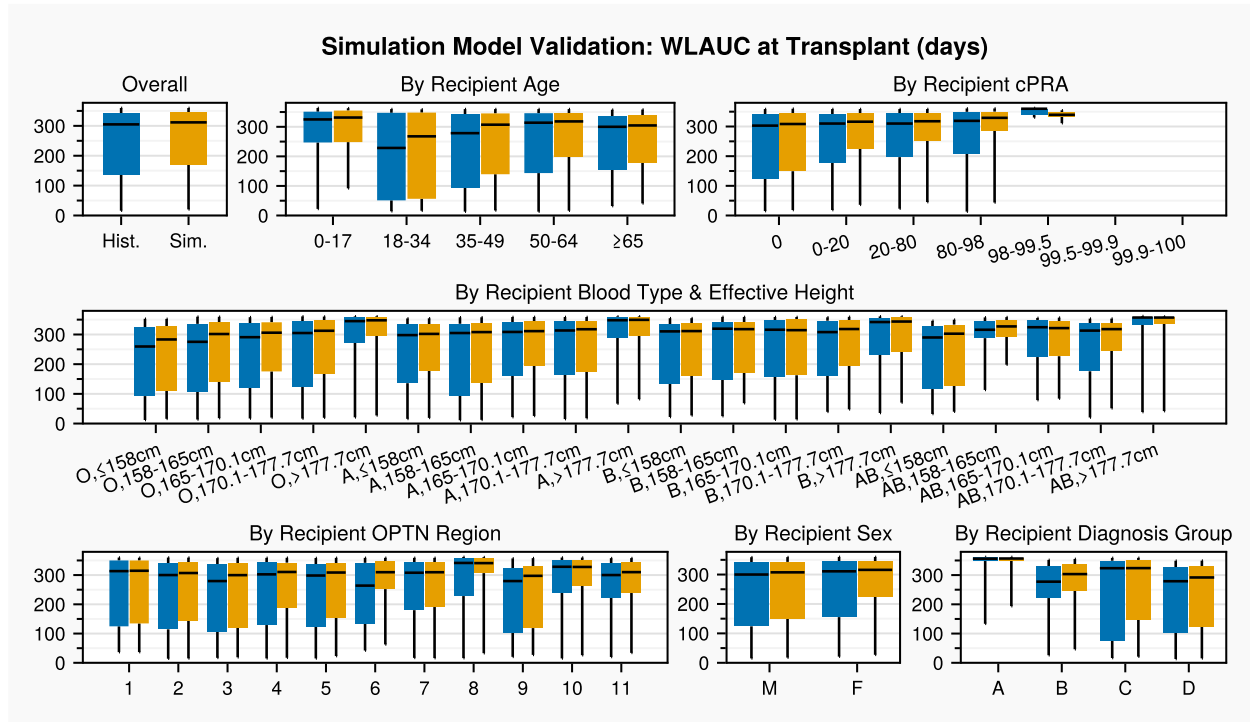


Figure S3. Comparison of WLAUC at transplant computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. No box plot is displayed if no patients within the associated group received a transplant. The whiskers run from the 5th to 95th percentile.

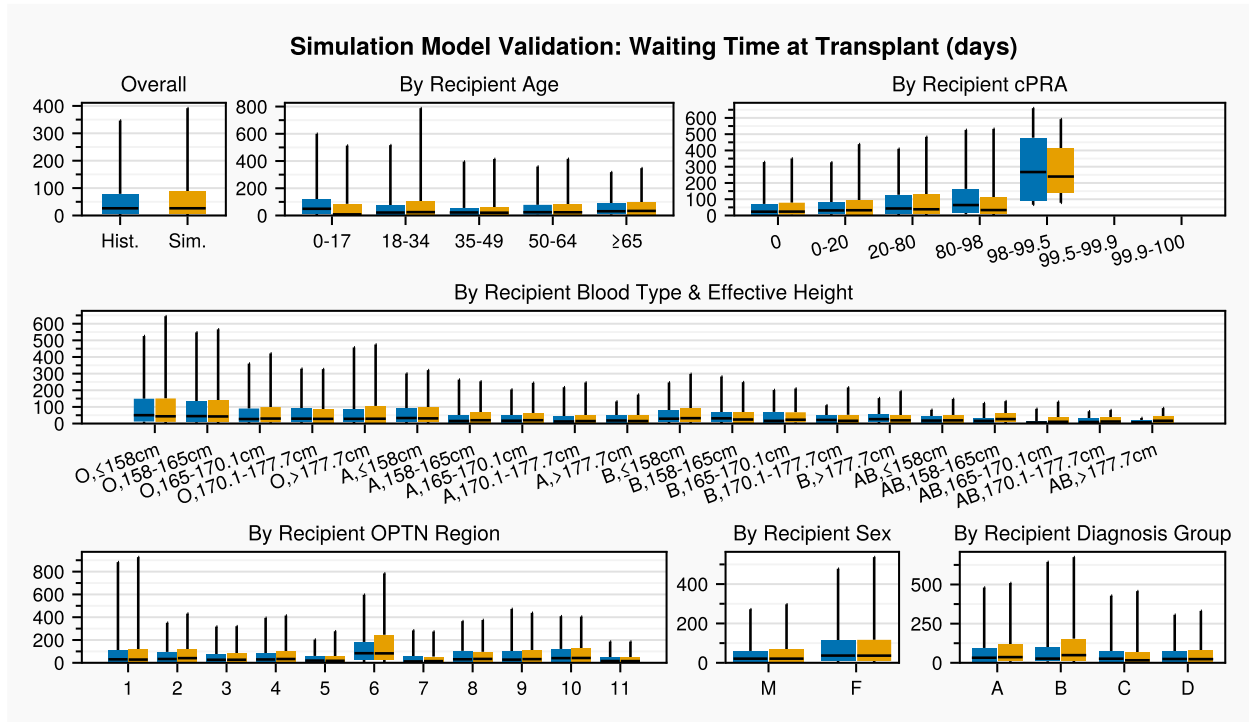


Figure S4. Comparison of waiting time at transplant computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. No box plot is displayed if no patients within the associated group received a transplant. The whiskers run from the 5th to 95th percentile.

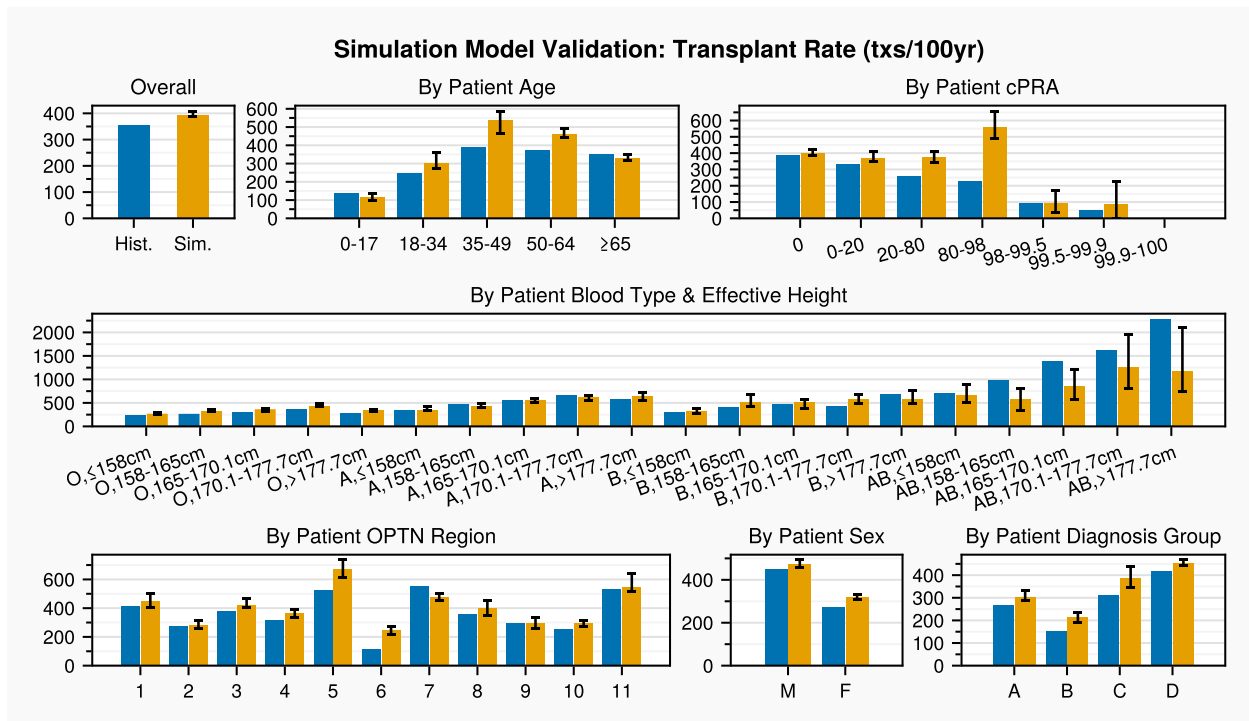


Figure S5. Comparison of transplant rate computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. The range bars run from the minimum to maximum simulated value.

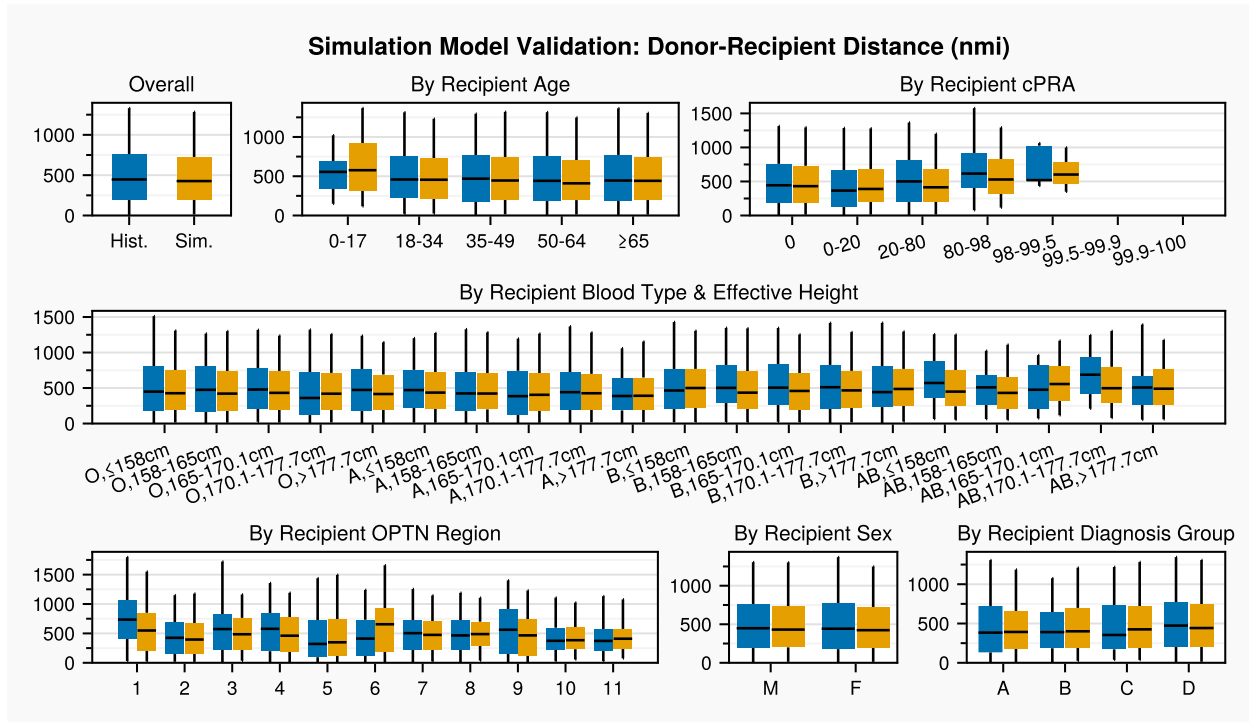


Figure S6. Comparison of donor-recipient distance computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. No box plot is displayed if no patients within that particular group received a transplant. The whiskers run from the 5th to 95th percentile.

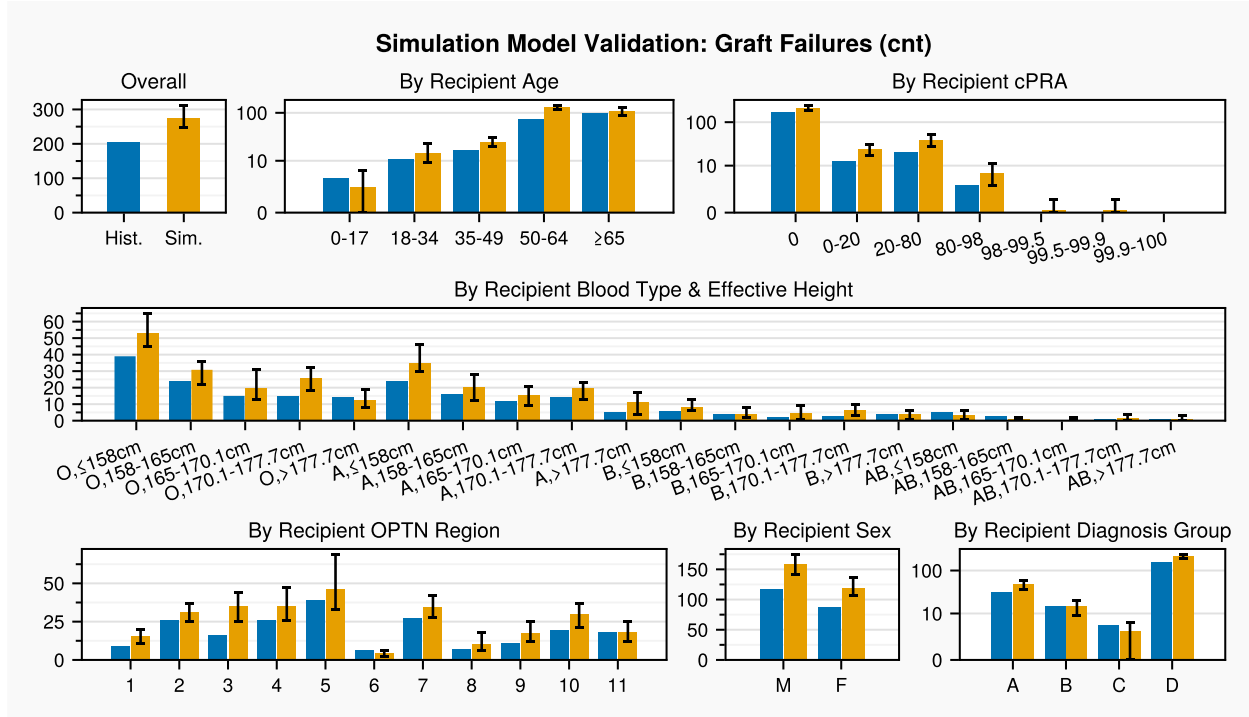


Figure S7. Comparison of graft failures computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. The range bars run from the minimum to maximum simulated value. Note that a psuedo-log-scale is used for the y-axis in some plots to be able to display large value ranges.

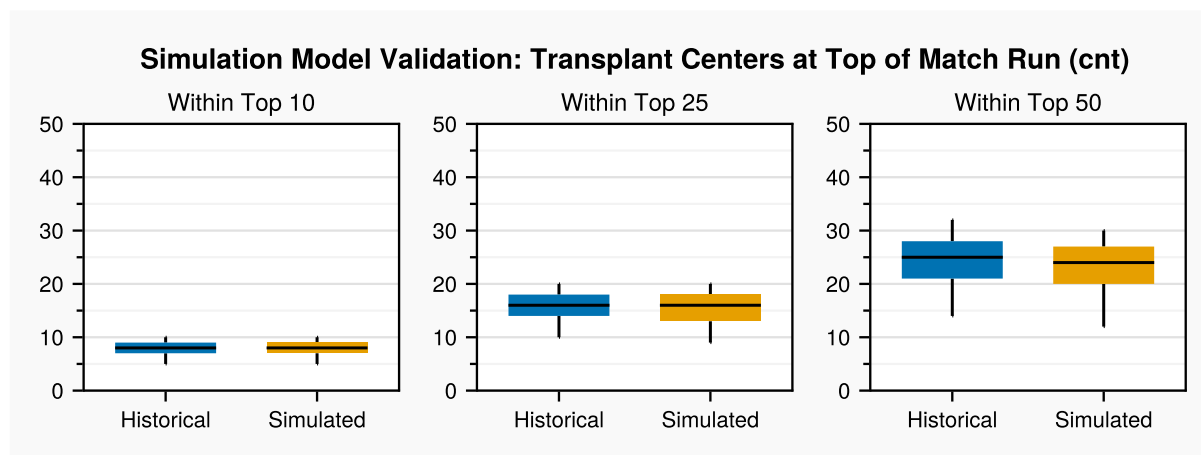


Figure S8. Comparison of transplant centers at top of match run computed from 2024 data (blue) and from 15 simulations of 2024 (yellow). Whiskers are drawn to the most extreme data point within 1.5 times the interquartile range.

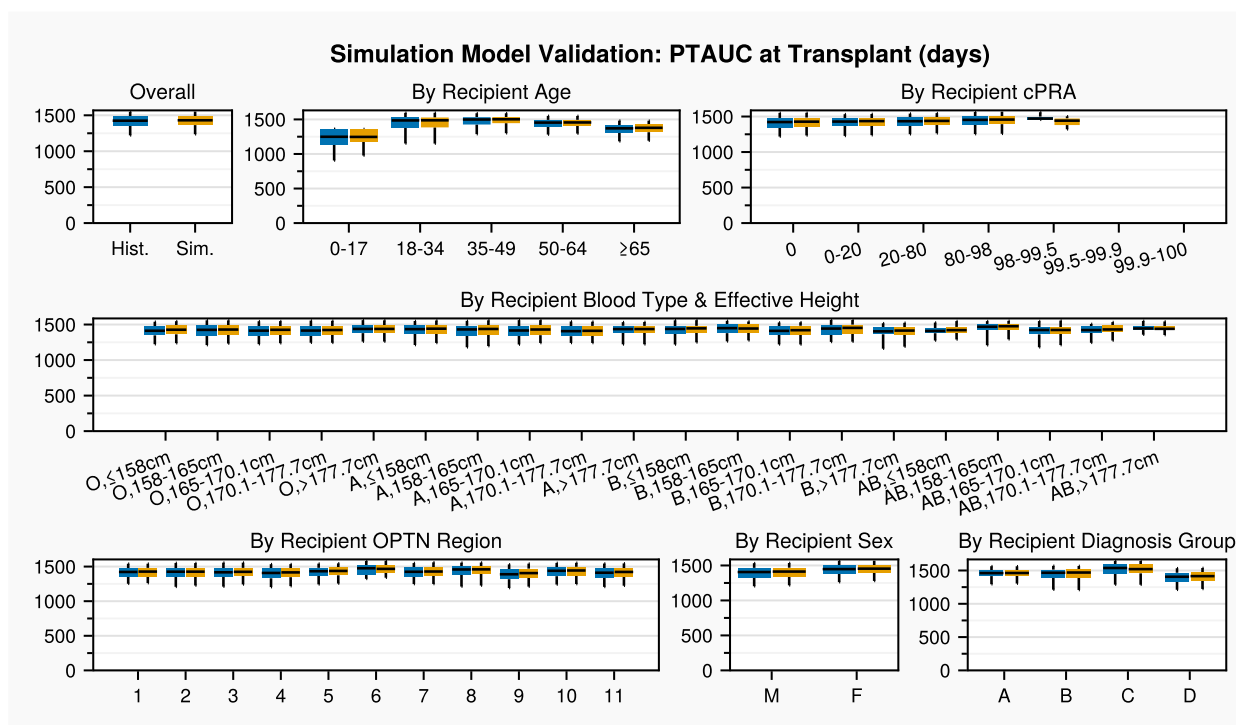


Figure S9. Comparison of PTAUC at transplant computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. No box plot is displayed if no patients within the associated group received a transplant. The whiskers run from the 5th to 95th percentile.

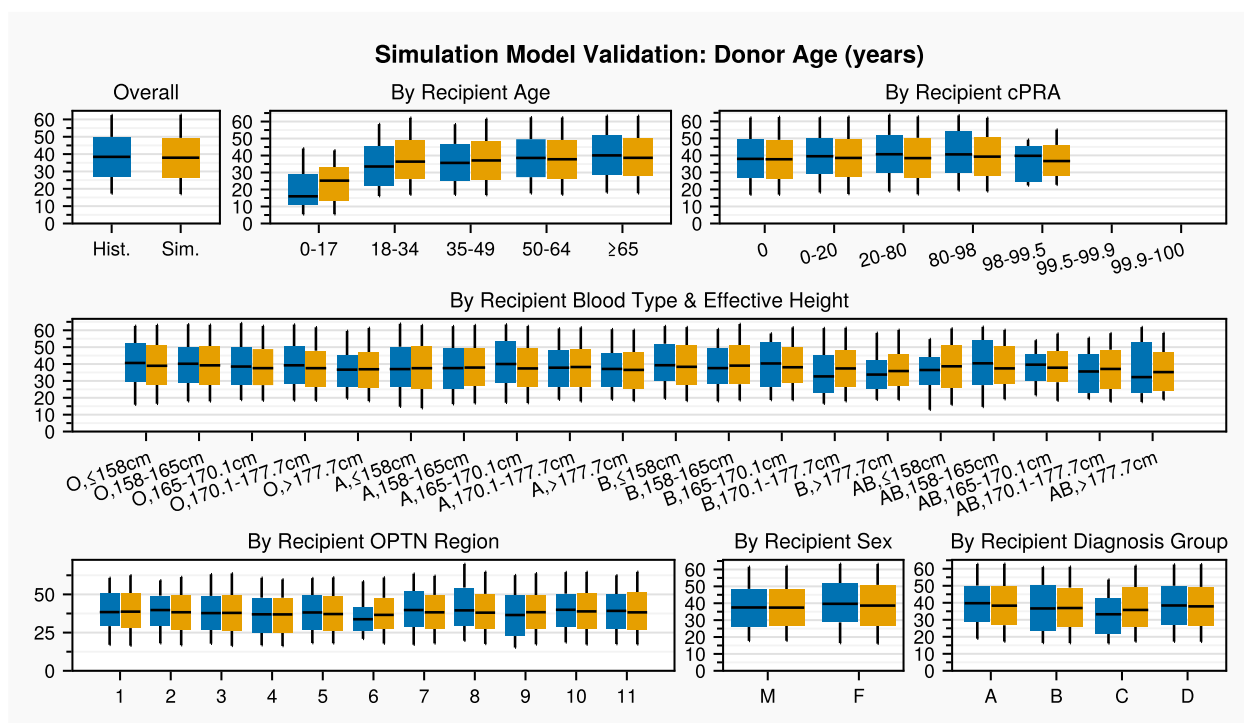


Figure S10. Comparison of donor age computed from 2024 data (blue) and from 15 simulations of 2024 (yellow), over all patients and select stratifications. No box plot is displayed if no patients within the associated group received a transplant. The whiskers run from the 5th to 95th percentile.